Pathway Innovation LLC.	Design Review	Doc No.: FM-0052
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Project No.	Product Name	Part No./ Rev	Review Phase	Review Date
00017	Hydrophilix Foley Catheter	N/A	1 (Planning)	7/12/2019

Subject of Review

#	Topic	Details	Reference Documents
1	Project Plan	Review of document	PL-0013 rev A
2	Regulatory Plan	Review of document	PL-0014 rev A
3	Risk Management Plan	Review of document	RM-0018 rev A
4	Review of Phase 1 Deliverables	Each document was reviewed and discussed. Project team in attendance has concluded that the deliverables are complete and satisfy the Design Control requirements in accordance with QP-007 rev D. It is agreed upon to close PHASE 1 and progress into PHASE 2.	Refer to #1 - #3.

Action Items

#	Description	Assigned To	Due Date
N/A	N/A	N/A	N/A

Reviewers

Functional Group	Required?	Name	Title	Signature	Date
Quality Assurance / Regulatory Affairs	☒	Aaron Rogers	QA Director	Aaron Rogers	07/12/19
R&D/ Engineering	☒	Dave Stroup	Managing Partner	[signature]	7/12/19
Manufacturing	☒	Jonas Cochran	Managing Partner	[signature]	7/12/19
Marketing	☐				
Independent Reviewer	☒	Tom Cassou	Operations / Business Development	[signature]	7/12/19
Other	☐				
Other/	☐				

1.0 Purpose

1.1 The purpose of this document is to provide details associated with design and development project for initial market release of the Hydrophilix (HDX) Foley Catheter. This project is documented under PRJ-0017.

2.0 Scope

2.1 The plan applies to the HDX Foley Catheter described below, which will be initial distributed in the US.

3.0 References

3.1 PRJ-0017 – Design Project, Hydrophilix Foley Catheter Product Introduction (US)

3.2 QP-007– Design Control

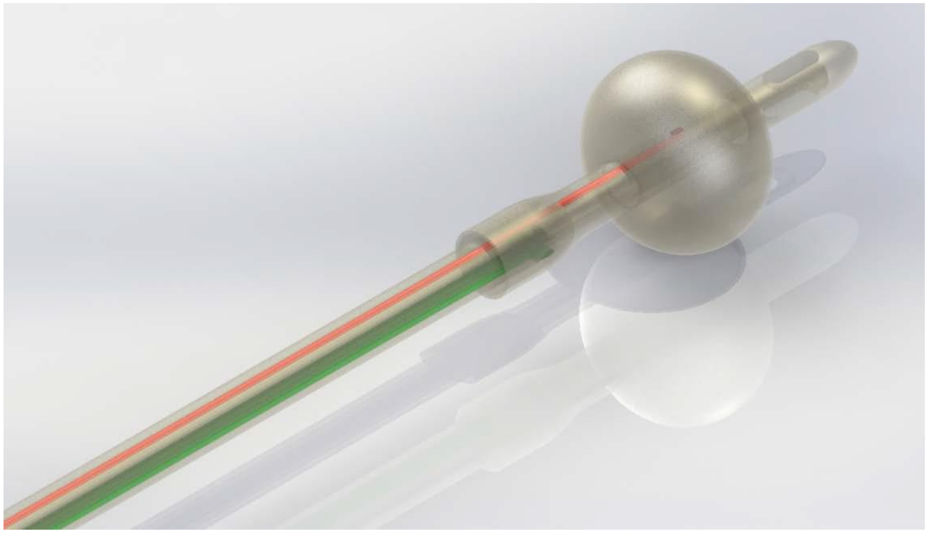
4.0 Background

4.1 Hydrophilix, Inc. was spun-out of the University of California, Los Angeles based on the Hydrophilix coating technology, a direct result of federal, state, and local support. The proof-of-concept began with basic research funding from the National Science Foundation Sustainable Chemistry Program and fellowships for individual scholars. The technology was further developed for commercial application, propelled with seed funding from the NSF Clean Energy for Green Industry, UCLA's Physical Sciences Entrepreneurship and Innovation Fund, and private investment.

4.2 When devices are coated with the Hydrophilix coating, a thin hydration layer prevents pathogens from adhering to the device surface and proliferating into biofilms. Because the coating does not kill microbes, the creation of superbugs is avoided. The hydrated coating is non-toxic and is applied in minutes.

5.0 Product Description

5.1 The HDX Foley catheter is a single use device that intended for use in the drainage and/or collection and/or measurement of urine. It is constructed of 2 lumens, which are for balloon inflation and drainage. The catheter will be constructed of medical grade silicone. The size offering will be 14, 16, and 18Fr with balloon volumes of 10ml – 30 ml. Refer to the following illustration of the device:



6.0 Product Configuration

6.1 The product will be offered in 3 sizes: 14, 16, and 18 Fr.

6.2 Devices will be packaged individually in a sterile barrier Tyvek/poly pouch, which 10 units will be incorporated per box.

7.0 Indication for Use Statement

7.1 Urological catheter intended for drainage/irrigation of urinary tract.

8.0 Goals & Objectives

8.1 Complete the following Design Control Activities defined in QP-007

8.1.1 Planning

8.1.2 Design Input

8.1.3 Design Output

8.1.4 Design Verification and Validation

8.1.5 Prepare and submit FDA pre-market notification (510k) application and gain clearance to distribute in the US.

8.1.6 Transfer the product to manufacturing to support commercialization of product.

9.0 Resources

9.1 Refer to the following table for the internal and external resources that are required to execute the design and development plan:

Internal/ External	Function	Name	Title
Internal	Project Leader / R&D	Dave Stroup	Managing Partner
Internal	R&D	Arthur Deptala	Managing Partner
Internal	Manufacturing	Jones Cochran	Managing Partner

Internal	Regulatory/ Quality	Aaron Rogers	QA Director
Internal	Independent Reviewer	Tom Cassou	Operations / Business Development
External	RA/ QA	Neal Hartman	Consultant

9.2 External testing services are provided by:

9.2.1 Biocompatibility: Nelson Labs

9.2.2 Sterilization: Steris

9.2.3 Transportation Testing: Westpak

9.3 Key Component and Process Suppliers:

9.3.1 Non-Sterile Foley Catheter: Ningbo Yingmed

9.3.2 Surface Treatment Solution: Hydrophilix, Inc.

9.3.3 Sterile Barrier Packaging: Steripax

9.3.4 Sterilization: Steris

10.0 Schedule/ Timelines

10.1 Schedules and timelines for each phase will be updated throughout the project and is maintained as part of the project module in Arena PLM.

11.0 Change History

Revision	Change Description
A	Initial Release

1.0 Purpose

1.1 The purpose of this document is to provide the details of a regulatory plan for initial market release of the Hydrophilix (HDX) Foley Catheter. This project is documented under PRJ-00017.

2.0 Scope

2.1 The plan applies to the applies to the HDX Foley Catheter described below, which will be initial distributed in the US.

3.0 References

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3.2 QP-007– Design Control

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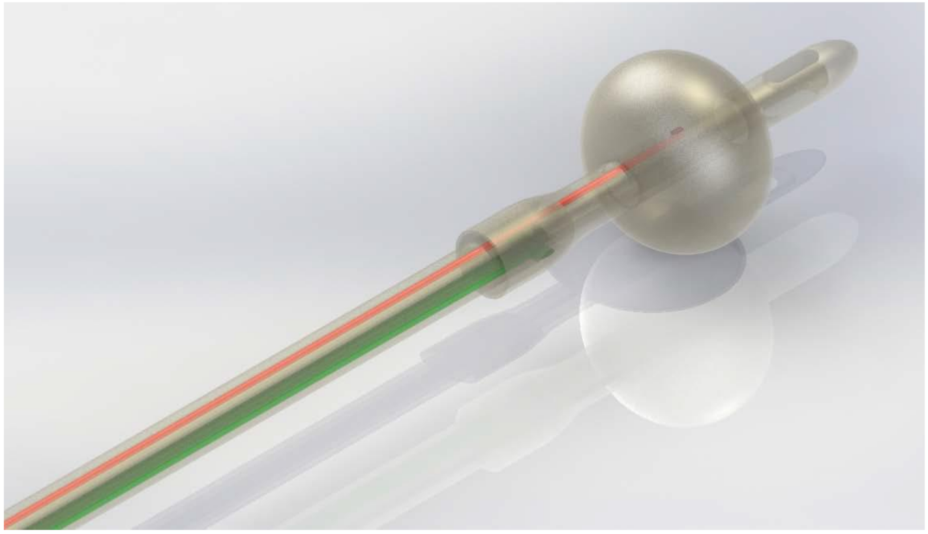

6.0 Commercialization/ Distribution Strategy

6.1 US Market

7.0 Regulatory Strategy

7.1 US Market

7.1.1 Device Classification: Class II

7.1.1.1 Product Code: EZL

7.1.1.2 Regulation Number: 876.5130

7.1.2 Facility Registration & Device Listing

7.1.3 Market Entry Requirements: Premarket Notification (510k)

7.1.4 Potential Predicate Devices

7.1.4.1 K091767 – Cook 3 Way Silicone Foley Balloon Catheter

7.1.4.2 K142636 – Medline Silicone Foley Catheter

7.1.4.3 K181616 – PSM Silicone Foley Catheter

8.0 Clinical Considerations

8.1 Indication for Use

8.1.1 Urological catheter intended for drainage/irrigation of urinary tract.

8.2 User Types

8.2.1 Pediatrics & Adults, Male & Female

8.3 User Environment

8.3.1 Healthcare Facility, Prescription Use

9.0 Associated Product Claims

9.1 There are no claims outside of the indication for use statement referenced in §8.1

10.0 Shelf-Life Considerations

10.1.1 A minimum shelf-life of one (1) year with the end goal of three (3) years.

11.0 Applicable Guidance Documents, Standards, and Regulations

11.1 Market Clearance

11.1.1 21 CFR 807 – Establishment registration and device listing for manufacturers and initial importers of devices

11.1.1.1 Subpart B – Procedure for Device Establishments

11.1.1.2 Subpart E – Premarket Notification Establishment

11.1.2 Guidance for the Content of Premarket Notifications for Conventional and Antimicrobial Foley Catheters

11.2 Quality Management System

11.2.1 21 CFR 820 – Quality System Regulations

11.2.2 21 CFR 803 – Medical Device Reporting

11.2.3 21 CFR 806 – Medical Devices; Reports of Corrections and Removals

11.3 Risk Management

11.3.1 ISO 14971:2012 – Medical Devices - Applications Of Risk Management To Medical Devices

11.4 Labeling

11.4.1 21 CFR 801 – Labeling

11.4.2 Guidance for Industry – Alternative to Certain Prescription Device Labeling Requirements – 1/21/2000

11.4.3 81 FR 38911 – Use of Symbols on Labeling

11.4.4 ANSI / AMII / ISO 15223-1:2016 – Medical devices — Symbols to be used with medical device labels, labeling and information to be supplied.

11.5 Biosafety

11.5.1 ANSI / AAMI / ISO 10993-1:2009 – Biological Evaluation of Medical Devices - Part 1: Evaluation and Testing Within a Risk Management Process

11.5.2 FDA-2013-D-0350, June 16, 2016 – FDA Guidance: Use of International Standard ISO 10993, “Biological Evaluation of Medical Devices Part 1: Evaluation and Testing within a risk management process”

11.6 Performance

11.6.1 ASTM F623: 2019 – Standard Performance Specification for Foley Catheter

11.6.2 ISO 80369-1: 2018 – Small-Bore Connectors For Liquids And Gases In Healthcare Applications - Part 1: General Requirements

11.7 Stability

11.7.1 ASTM F1980-16 – Standard Guide for Accelerated Aging of Sterile Barrier Systems for Medical Devices

11.8 Sterilization

11.8.1 ANSI / AAMI / ISO 11135:2014 - Sterilization of health care products – Ethylene Oxide – Requirements for development, validation and routine control of a sterilization process for medical devices

11.9 Packaging

11.9.1 ISO 11607-1:2019 – Packaging for Terminally Sterilized Medical Devices - Part 1: Requirements for Materials, Sterile Barrier Systems and Packaging Systems

11.9.2 ASTM D4332-14 – Standard Practice for Conditioning Containers, Packages, Or Packaging Components for Testing

11.9.3 ASTM D4169-16 – Standard Practice for Performance Testing of Shipping Containers and Systems.

12.0 Change History

Revision	Change Description
A	Initial Release

1.0 Purpose

The purpose of this document is to indicate the requirements of the risk management plan that is associated with the development of the Hydrophilix (HDX) Foley Catheter.

2.0 Scope

2.1 The document applies for the design project associated with the product introduction of the HDX Foley Catheter.

2.2 This document will identify the activities planned for risk management.

3.0 References

3.1 ISO 14971:2012 – Medical devices – Application of risk management to medical devices

3.2 QP-002 – Document Control

3.3 QP-004 – Risk Management

3.4 QP-007 – Design Control

3.5 QP-011 – Post Market Surveillance

3.6 Project#00017 – Hydrophilix Foley Catheter product introduction (US)

4.0 Responsibilities & Authorities

4.1 The risk management team will include members of Pathway. Team members will include individuals representing Research & Development (R&D), Manufacturing, Marketing, Regulatory Affairs (RA), and Quality Assurance (QA).

Individuals that participate in risk management activities will have knowledge of and experience with the device (or similar device) and how it is manufactured.

4.2 The risk management team will be required to complete all activities indicated in this document. Team members from Pathway will be required to review all content of risk management deliverables and agree on its conclusions with documented approval.

5.0 Product Description

The HDX Foley Catheter is intended for use to drain/irrigate the urinary tract.

6.0 Product Life-Cycle Phases

6.1 Design

In the appropriate phase of the design control process, per QP-007, a Hazard Analysis (HA) and Design Failure Mode Effects Analysis (DFMEA) will be developed per, QP-004, as input to the design of the device.

Criteria of risk acceptability is identified in QP-004.

If additional risk(s) are identified during the progression through the design control phases, especially in verification and validations, risk management deliverables will be reviewed and updates appropriately.

6.2 Process Validation

In the “Transfer to Manufacturing” phase of the design control process per QP-007, a Process Failure Mode Effects Analysis (PFMEA) will be developed as a deliverable process validation. Criteria of risk acceptability is identified in QP-004.

6.3 Production & Post-Production

Information collected from production and post-production activities will be evaluated for risk as part of the post-market surveillance process per QP-011. Risk management report will be review and newly identified risk or changes to existing risk will be incorporated.

Revised information to the risk management reports will be reviewed with R&D, RA, and QA to determine if action shall be taken to the product design to mitigate identified risk. Action shall be incorporated in a change control project per QP-007.

7.0 Verification Activities

Activities associated for risk controls (initial & mitigation) will be documented in the risk management reports.

8.0 Risk Management File

The risk management file consists of this plan and the risk management report (i.e., Hazard Analysis, DFMEA, & PFMEA). The risk management plan will be maintained as a controlled document per QP-002 and associated in the product’s Design History File (DHF).

9.0 Change History

Revision	Change Description
A	Initial release